Flow charts and operations

1 The flow chart shows operations being performed on s:

$$s \xrightarrow{-7} \square \xrightarrow{\times 6} \square$$

Complete the flow chart.

2 The flow chart shows operations being performed on t:

$$t \xrightarrow{+3} \square \xrightarrow{\div 5} \square$$

Complete the flow chart.

3 Complete the following flow charts to form new expressions:

a $y \xrightarrow{\times 5} \square \xrightarrow{+8} \square$

b $k \xrightarrow{\div 3} \square \xrightarrow{-7} \square$

c $p \xrightarrow{+15} \square \xrightarrow{\times 8} \square$

d $n \xrightarrow{+5} \square \xrightarrow{\div 3} \square$

e $x \xrightarrow{\times 4} \square \xrightarrow{+3} \square$

f $j \xrightarrow{\div 9} \square \xrightarrow{-7} \square$

g $p \xrightarrow{+28} \square \xrightarrow{\times 3} \square$

h $s \xrightarrow{-6} \square \xrightarrow{\times 7} \square$

Operations in reverse order

4 Complete the following flow charts to backtrack to the pronumeral in each expression:

a $6y + 9 \xrightarrow{-9} \square \xrightarrow{\div 6} \square$

b $5(q + 3) \xrightarrow{\div 5} \square \xrightarrow{-3} \square$

c $6(s - 9) \xrightarrow{\div 6} \square \xrightarrow{+9} \square$

d $\dfrac{n + 8}{5} \xrightarrow{\times 5} \square \xrightarrow{-8} \square$

e $\dfrac{j}{5} - 6 \xrightarrow{+6} \square \xrightarrow{\times 5} \square$

f $8y + 9 \xrightarrow{-9} \square \xrightarrow{\div 8} \square$

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5 Complete the following flow charts by identifying the required operations:

a $4(q+9) \xrightarrow{\square} q+9 \xrightarrow{\square} q$

b $5(r-7) \xrightarrow{\square} r-7 \xrightarrow{\square} r$

c $\frac{n+6}{9} \xrightarrow{\square} n+6 \xrightarrow{\square} n$

d $\frac{j}{2}-7 \xrightarrow{\square} \frac{j}{2} \xrightarrow{\square} j$

Two-step equations

6 Consider the equation $4x+16=24$.

a Identify which operations should be done to make x the subject of the equation:

$$4x+16 \xrightarrow{\text{Operation 1}} 4x \xrightarrow{\text{Operation 2}} x$$

b Complete the following flow chart by applying the same operations to the right-hand side of the equation:

$$24 \xrightarrow{\text{Operation 1}} \square \xrightarrow{\text{Operation 2}} \square$$

c Hence, find the value of x that will make the equation true.

7 Consider the equation $3(p+11)=57$.

a Identify which operations should be done to make p the subject of the equation:

$$3(p+11) \xrightarrow{\text{Operation 1}} p+11 \xrightarrow{\text{Operation 2}} p$$

b Complete the following flow chart by applying the same operations to the right-hand side of the equation:

$$57 \xrightarrow{\text{Operation 1}} \square \xrightarrow{\text{Operation 2}} \square$$

c Hence, find the value of p that will make the equation true.

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8 Consider the equation $\frac{j}{7} - 11 = -7$.

a Identify which operations should be done to make s the subject of the equation:

$$\frac{j}{7} - 11 \quad \xrightarrow{\text{Operation 1}} \quad \frac{j}{7} \quad \xrightarrow{\text{Operation 2}} \quad j$$

b Complete the following flow chart by applying the same operations to the right-hand side of the equation:

$$-7 \quad \xrightarrow{\text{Operation 1}} \quad \square \quad \xrightarrow{\text{Operation 2}} \quad \square$$

c Hence, find the value of s that will make the equation true.

9 For each of the following equations:

i Describe the operations required to make the pronumeral the subject.

ii Hence, solve the equation.

a $7p + 6 = 41$

b $2x - 5 = 11$

c $5(n - 15) = 35$

d $10(p + 10) = 120$

e $\frac{y}{8} + 3 = 5$

f $\frac{x}{3} - 11 = 9$

g $\frac{u + 22}{6} = 4$

h $\frac{y - 10}{2} = 5$

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